TEXORA · LECTURE SHEET

Apparel Manufacturing Management

How a garment factory is organised and run: the management functions applied to an order-driven factory, the department structure, the order cycle from enquiry to shipment, and the KPI arithmetic by which managers steer production.

AM-4201 · ADVANCED APPAREL MANUFACTURING-II · TOPIC 1 · LEVEL 4 TERM 2

1. Management in an order-driven factory

Apparel manufacturing is **make-to-order production**: nothing is sewn speculatively, and every order arrives with its own style, sizes, colours, price, delivery date and quality standard. Management is therefore the discipline of converting a stream of unlike, deadline-fixed orders into a stable factory routine. The classical functions map directly: **planning** (which line sews what, when — Topic 4), **organising** (departments and lines below), **staffing** (operator skill matrices matched to operations), **directing** (line supervision, hourly targets) and **controlling** (efficiency, quality and shipment KPIs, §4). The apparel-specific pressure is that the product changes faster than the factory: a line balanced for shirts meets trousers next week, so apparel management prizes **changeover speed and flexibility** over the long-run stability of classic mass production.

2. Department structure

DEPARTMENT	RESPONSIBILITY	KEY OUTPUTS / RECORDS
Merchandising	Buyer interface: costing, order confirmation, material booking, approvals, shipment follow-up.	Cost sheet, tech pack, time-and-action (T&A) calendar.
Sample / product development	Fit, PP and production samples; pattern and marker masters (with CAD, AM-4102 Study 4).	Approved samples, sealed standards.
Production planning & IE	Capacity planning, line loading, SMV database, line balancing, efficiency reporting (Topics 3–4).	Production plan, line layouts, efficiency reports.
Cutting	Fabric issue, spreading, cutting, bundling (Topic 5).	Cut panels, bundle tickets, cutting reports.

Sewing	The assembly lines; the factory's largest workforce and cost centre.	Hourly output, WIP, in-line quality data.
Washing / finishing	Wash processes (AM-4102), trimming, pressing, final check, packing.	Wash standards, audit results, packing lists.
Quality	Incoming, in-line and final audit systems (TE-3107, AM-4102 Study 3).	DHU charts, AQL audit reports.
Store / materials	Fabric and trims inventory, issue against consumption (Topic 6).	Stock ledger, issue records.
Commercial / HR / compliance	Export documentation, payroll, buyer social & technical compliance.	Shipment documents, audit certificates.

3. The order cycle

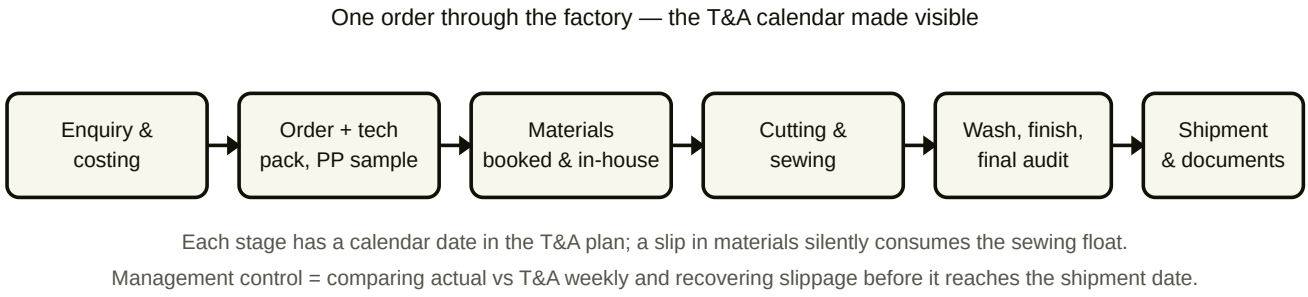


Figure 1 — The order cycle. The factory is managed as a chain of dated commitments, not as separate departments.

4. Steering numbers: capacity and efficiency

WORKED EXAMPLES — VERIFY EVERY NUMBER

(a) Daily line output. A line of 28 operators works 480 min/day at 60% efficiency on a garment of SMV 15 min. Attended minutes = 28 × 480 = 13,440. Earned minutes available = 13,440 × 0.60 = 8,064. Output = 8,064 ÷ 15 = **537.6 ≈ 538 garments/day**. This one formula — output = (operators × minutes × efficiency) ÷ SMV — is the factory's master equation: every management topic in this course adjusts one of its four terms.

(b) Efficiency from production data. A line of 40 operators produces 800 garments of SMV 12.0 min in a 480-min day. Earned minutes = 800 × 12.0 = 9,600. Attended = 40 × 480 = 19,200. Efficiency = 9,600 ÷ 19,200 × 100 = **50.0%**.

(c) The KPI set. Efficiency (above) · **DHU** at end-of-line (target < 5, AM-4102 Study 3) · **on-time shipment %** · **order lead time** (order date → ex-factory) · **cut-to-ship ratio** (shipped ÷ cut;

losses hide here) · **cost per minute** (factory cost ÷ attended minutes) against the CM price earned per minute. A dashboard of six honest numbers beats a wall of reports.

5. Management problems peculiar to apparel

- **Style changeovers:** every new style resets line balance and operator learning; management answers with pre-production meetings, pilot runs and feed planning (Topic 4).
- **Labour intensity:** sewing resists automation; workforce skill, absenteeism and retention are production variables, managed by training matrices and multi-skill operators.
- **Material dominance:** fabric and trims are 60–70% of garment cost; material delays and consumption overruns (Topic 6) outweigh most labour savings.
- **Compliance as licence:** buyer audits on safety, labour and environment gate the order book; compliance is a management system, not a department's hobby.

Glossary

TERM	MEANING
SMV	Standard minute value: the standard time allowed for an operation or garment, including allowances (Topic 3).
T&A calendar	Time-and-action plan: dated milestones for one order from confirmation to shipment.
Earned minutes	Output × SMV — the standard minutes a line actually produced.
Cut-to-ship ratio	Garments shipped ÷ garments cut; exposes loss, rejection and theft between cutting and dispatch.
CM	Cut-and-make price: the conversion price the factory earns, excluding materials.

Key points and summary

- Apparel management converts unlike, deadline-fixed orders into routine: plan, organise, staff, direct, control.
- The order cycle is managed as one dated chain (T&A); slippage is recovered upstream, not at shipment.
- Master equation: output = operators × minutes × efficiency ÷ SMV. Efficiency = earned ÷ attended minutes.
- Apparel's special problems: changeovers, labour intensity, material cost dominance and compliance.

Practice

Answer in your own words; brief model answers follow.

1. A 35-operator line at 55% efficiency sews a garment of SMV 18 min in a 480-min day. Calculate the daily output.
2. A line produces 950 garments of SMV 10.5 min with 42 operators in 480 min. Calculate its efficiency.
3. Fabric for an order arrives 6 days late. Using the T&A idea, state two recovery actions that do not move the shipment date, and one risk of each.
4. Why is the cut-to-ship ratio a management KPI and not just a store record?
5. Which term of the master equation does a training programme change, and which does a better marker change?

Answer key

1. Attended = 35 × 480 = 16,800 min. Earned available = 16,800 × 0.55 = 9,240. Output = 9,240 ÷ 18 = 513 garments/day.
2. Earned = 950 × 10.5 = 9,975 min. Attended = 42 × 480 = 20,160. Efficiency = 9,975 ÷ 20,160 × 100 = 49.5%.
3. Examples: overlap stages — start cutting approved colours/sizes first and resequence the line plan (risk: partial-lot confusion, size shortages at packing); add overtime or a parallel line for the first days (risk: cost, fatigue-driven quality drop). Both spend money or complexity to buy back calendar; both are cheaper than air freight or a missed vessel.
4. Because every garment cut but not shipped consumed fabric, labour and capacity without earning: the gap localises loss — wash rejection, audit failure, damage, pilferage — and each cause belongs to a different department. A ratio drifting below ~0.98 is an investigation trigger.
5. Training raises efficiency (and can lower effective SMV only via method, Topic 3 — SMV itself is a standard, not a hope). A better marker changes none of the four terms directly; it cuts material consumption per garment (Topic 5/6 economics), which is cost, not line output.